

Chapter 18: The Probabilistic Method and Random Graph

18.1 The Original Result

The original use of the probabilistic method was to provide a lower bound for Ramsey numbers, namely that $R(m, m) > 2^{m/2}$.

ERDŐS. *There exists a graph G on $N = 2^{m/2}$ vertices that has neither an independent set nor a clique of size m .*

PROOF. Start with N vertices. For each pair of vertices, toss an unbiased independent coin to see whether they are joined by an edge or not. The result is a graph G_N . We want to calculate the probability that G_N has the desired property.

The probability that any fixed set A of m vertices forms a clique is $\frac{1}{2^{\binom{m}{2}}}$, since every edge must be present. The probability that A forms an independent set is the same.

Thus, the probability that some subset of size m forms a clique or independent set, is at most $\binom{N}{m}$ times this quantity. That is,

$$\Pr(\exists \text{ bad } A) \leq (\# \text{ possible } A) \times \Pr(\text{specific } A \text{ is bad}) = \binom{N}{m} \times \frac{2}{2^{\binom{m}{2}}}.$$

Now, by a bit of arithmetic,

$$\binom{N}{m} \times \frac{2}{2^{\binom{m}{2}}} < \frac{2N^m}{m! 2^{m(m-1)/2}} < \frac{N^m}{2^{m^2/2}} = \left(\frac{N}{2^{m/2}}\right)^m = 1.$$

So,

$$\Pr(\exists \text{ bad } A) < 1.$$

Or put the other way around, the probability that there is no bad A is nonzero: one of these random graphs has neither an independent set nor a clique of size m . Thus there exists a graph on N vertices with the desired property. $\square$

Note that a probabilistic proof like this can also be thought of as a counting argument. We count all the graphs on N vertices. Then we count all the graphs on N vertices with a large clique or large independent set, and find that they cannot account for all the graphs.

18.2 Linearity of Expectation

Suppose set X is constructed by some random process. Suppose each element x is in X with probability π_x . Then, even though whether one x is in X might depend heavily on whether another x' is in X or not, by linearity of expectation the expected size of X is $\sum_x \pi_x$.

As an application:

THEOREM. *For any graph G there exists a partition of the vertices of G into two sets L and R such that at least $1/2$ the edges go between L and R .*

PROOF. Construct L and R at random by putting each vertex independently in L with probability $1/2$. The probability that a given edge goes between L and R is $1/2$ (Why?). So the expected number of edges between L and R is half the edges. Thus, there exists a partition where the number of edges is at most half the total and one where the number of edges is at least half the total. $\square$

Actually, it is not hard to explicitly construct such L and R .

THEOREM. *The domination number of an r -regular graph of order n is at most*

$$\gamma \leq \frac{1 + \ln r}{r} n.$$

PROOF. Constructing a set Y at random by taking each vertex independently with probability $p = \ln r/r$. Then create dominating set D by taking Y and adding every vertex that is not dominated by Y .

The probability a vertex v is in D but not Y is the probability that no vertex in $N[v]$ is in Y . This equals $(1 - p)^{1+r} \approx 1/r$. Hence the expected size of D is the value in the theorem. $\square$

Alon showed that there are graphs where the domination number is near this value. The proof also uses probabilistic methods.

18.3 Turán's Theorem

Recall:

THEOREM. *The maximum number of edges in a graph on n vertices without a k -clique is $n^2(k-2)/(2(k-1))$.*

Rephrased it reads:

THEOREM. *If G is a graph with n vertices and q edges, then it has a clique of size at least $n^2/(n^2 - 2q)$.*

PROOF. We start by showing a more general statement:

$$\text{clique-size} \geq \sum_{v \in V} \frac{1}{n - \deg v}.$$

Choose a random ordering π of the vertices. For a given ordering, let C_π be the set of all x such that x is adjacent to every vertex before it in the ordering π . This is a clique, as every vertex added to C_π is adjacent to all the earlier vertices of C_π .

Let us compute the expected size of C_π . Consider a vertex v and the set Z_v consisting of v and its non-neighbors. The probability p_v that $v \in C_\pi$ equals the probability that, out of all the vertices in Z_v , vertex v appears first in π . Hence this probability $p_v = 1/|Z_v|$.

By linearity of expectation it follows that the expected size of C_π equals $\sum 1/(n - \deg v)$. And there must be an actual clique with size at least the average. This proves the above claim.

Now, by calculus it follows that $\sum 1/(n - \deg v)$ for fixed q is minimized when the degrees are equal. So this sum is at least $n \times 1/(n - 2q/n)$. Hence there is a clique of this size. $\square$

18.4 Sperner's Theorem

We say that two sets are *incomparable* if neither is a subset of the other. A collection $\mathcal{S}$ of sets is called an antichain or a *Sperner family* if every pair of sets in $\mathcal{S}$ are incomparable. One can use probabilistic methods to find an upper bound on the size of a Sperner family:

THEOREM. Let $\mathcal{S}$ be a Sperner family of subsets of $\{1, \dots, n\}$. Then $|\mathcal{S}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

PROOF. We define a **chain** of sets as a sequence of subsets such that the next subset is obtained from the previous subset by adding one element.

Construct a chain $\mathcal{C}$ starting at $\emptyset$ and ending at $\{1, \dots, n\}$ at random. This means that at each stage we pick the next element to add to the set uniformly at random.

Let E denote the expected value of $|\mathcal{C} \cap \mathcal{S}|$. Then $E \leq 1$, because $|\mathcal{C} \cap \mathcal{S}| \leq 1$ for any chain $\mathcal{C}$. On the other hand, the probability that any particular set S of cardinality s is in $\mathcal{C}$ is $1/\binom{n}{s}$. (Why?) In particular, this probability is always at least $1/\binom{n}{\lfloor n/2 \rfloor}$. So

$$1 \geq E \geq |\mathcal{S}| / \binom{n}{\lfloor n/2 \rfloor},$$

which gives the bound. $\square$

Actually, the above proof can be used to prove the LYM inequality that extends Sperner's theorem.

18.5 Properties of the Random Graph

Let $G(n)$ be the graph on n vertices with edges present independently at random with probability $1/2$. We say a property holds **almost surely** if its probability of holding goes to 1 as n goes to infinity.

THEOREM. Almost surely $G(n)$ has diameter 2.

PROOF. Well, there is a vanishingly tiny probability of diameter 1. For vertices x and y , the probability that x and y have z as a common neighbor is $1/4$. The probability they have no common neighbor is therefore $(3/4)^{n-2}$. Thus the probability that there exists a pair not at distance two is at most n^2 times this probability, which is vanishing small. $\square$

THEOREM. The independence number of $G(n)$ is almost surely $2 \log_2 n + O(1)$.

Go back to the proof earlier. We argued that if $n = 2^{-k/2}$, then the probability that $G(n)$ does not have an independent set of size k was nonzero. That is, if $k = 2 \log_2 n$, then the probability that the independence number is less than k is nonzero. However, to show that it tends to 1 takes more work, in particular, bounds from probability theory such as Janson's inequality.

Bollobás showed:

THEOREM. *The chromatic number of $G(n)$ is almost surely $n/(2 \log_2 n)$.*

Note that the result on independence number gives the lower bound on the chromatic number, but it does not give an upper bound. Because, if we try to remove a maximum independent set and recurse, what remains is not a random graph.

Other facts that hold almost surely:

- the minimum degree is $n/2 - \sqrt{(n \log n)/2} (1 + o(1))$ with unique vertex of minimum degree.
- the connectivity equals the minimum degree.
- there is a perfect matching (up to one vertex) and indeed a hamilton cycle.
- there is no nontrivial automorphism.

There is a cute result about the infinite random graph:

THEOREM. *All random graphs with a countably infinite set of vertices are isomorphic.*

PROOF. Let $v_1, v_2, \dots$ be any ordering of the vertices of G_1 . Construct a bijection as follows. Pick a vertex w_0 in G_2 . Pick a neighbor or non-neighbor w_1 in G_2 depending on whether v_0 and v_1 are adjacent.

In general, let w_i be any vertex in G_2 such that bijection $v_j \rightarrow w_j$ for $1 \leq j \leq i$ is an isomorphism. Always exists because $1/2^{i-1}$ vertices in G_2 are candidates.

□

Other models of random graphs include random regular graphs and power law graphs.

18.6 Evolution of the Random Graph

One can generalize $G(n)$ to $G(n, p)$, which is the random graph on n vertices where every edge is present independently at random with probability p . It is useful to think of starting with $p = 0$ and steadily increasing p . It turns out that there is a “phase change” around $p = 1/n$.

If $p \gg 1/n$, then almost surely the graph is connected, and indeed hamiltonian, and many of the properties that we know for $G(n, 1/2)$ hold. Actually, one doesn’t have to go much above $1/n$. For example, the threshold for no vertex being isolated is around $p = \ln n/n$, and similarly for connectivity.

In contrast, if $p \ll 1/n$, then almost surely all components are trees, all of size at most $O(\log n)$. Cycles only appear when p is around $1/n$. Indeed, it turns out that the threshold for the appearance of a subgraph H is determined by its densest subgraph. Let $m(H)$ be the *maximum average degree* of a subgraph of H : that is, the maximum ratio $2s/r$ taken over all subgraphs of H on r vertices and s edges. Then the threshold is $p = n^{-2/m(H)}$.

Hence at $p = 1/n$, only unicyclic components are added. However, by $p = 2/n$ it turns out that almost surely there is one giant component that contains almost all the vertices. So the phase change happens at exactly $1/n$. Indeed, it can be proven that many of the thresholds are very abrupt.